SIMATS ENGINEERING

ASSESSMENTTOOL3:TechnicalPresentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloudcomputing technologies including hardware-software evolution, virtualization, web services,and servicemodels to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I A.Sunitha (192571079)(Name / Reg No) certify that this submission is my original work and that I have adhered totheguidelines specified for this assessment.

Iunderstand thatanyviolation of academic integrity rules will result in disciplinary action.

Signature of thestudent: A.Sunitha.

Assessment Tasks

Technical Presentation Topics

Preparea6-8 minutepresentation on any one of the following topics. Your slides must include definitions, architecture, industryrelevance,and one practical example.

Here are 10 additionaltechnicalpresentation topics in a similar style, suitable for a 6–8 minute presentation withdefinitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - o Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - o Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - o Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - o Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - o Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - o Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
   - o Discuss AI/ML cloud services, architecture, and industry applications.

Containerization and cloud-Native Applications: Transforming Modern Software Deployment of Explore containers, container architecture, benefits and real-world deployment Example:

1. Introduction

Containerization is a technology used to package an application along with its code, libraries and dependencies into a single unit called a container.

It helps applications run consistently on different computers and cloud platforms.

Example: An application bi developed on a laptop can run on a cloud server without many compatability problems.

2. Aim and objectives

Aim:- To understand containerization supports the development, deployment and management of modern cloud-native applications.

Objectives:

→ understand the concept of containerization

→ Learn the basic architecture of containers.

when demand increases.

- Application isolation: Applications run in separate environments, reducing interference.

5. Container Architecture:

1. Application: The software that needs to run.
2. container Image: containers the application and its dependencies.
3. container Runtime: Runs the container.
4. cloud Infrastructure: provides computing resources such as servers and storage.
5. Kubernetes: Manages containers and help Scale applications.

Example:

Docker creates the containes, the containes runtime runs it and kubernetes manages multiple containers in the cloud.

7. Real-world use case:-

Netflix

Netflix provides streaming services to millions of users. container-based technologies can help applications run as independent services, allowing teams to deploy and update different services independently.

How containerization helps:-

- services can be updated independently.
- Applications can be deployed quickly
- Resources can be increased when more users watch videos.
- Resources can be reduced when demand decreases.

Note:- These are general benefits of containerization in streaming systems; the ppt does not establish that every Netflix service uses containers.

8. Conclusion:

containerization is a important technolog
in modern cloud computing. It makes applica
dep portable, light weight and easier to deploy.
Docker helps create containers while kubernetes
helps manage them. Together with cloud
infracture these technologies support the
development and duelopment of and delivery
of modern applications.

SIMATS ENGINEERING SIMATS

CSA1520-Cloud Computing & Big Data Analysis

Containerization and cloud-Native Applications:Transforming Modern Software Deployment Of Explore Containers,Container architecture,benefits,and a real-world deployment example

Presented by :

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GPS Map Camera

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Lat 13.02642° Long 80.017045°

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Google

Minimum slide flow:

- Title andobjective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria Technical	**Weightage** 30%	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Content		Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Contentis inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism